



Clouded Vision

Atmospheric dynamics remain very poorly understood. Models that deem to predict rainfall patterns can at best be called a joke.

After a scorcher last year, when India faced the worst drought in three decades, the announcement by the India Meteorological Department (IMD) that the monsoon would be 'normal' this year may come as a relief. So should we applaud them for this much-needed reprieve or look deeper as something seems amiss in their 'operational' long-range monsoon forecasting technique? Something is indeed very wrong.

In the last two decades the IMD has invariably only forecast 'normal' monsoons. Despite the huge variations, India has witnessed bad years like the floods of 1994, the droughts of 1987, 2002, 2004 and 2009. Some suggest that monsoon forecasting is fast turning into a joke.

In a moment of rare scientific honesty, Ajit Tyagi, the director-general of the IMD, has now admitted that 'the extremes are really difficult to forecast'. To put this in perspective, the Indian weather office has never ever been able to predict a drought in the last 130 years. Look at the basic statistics in the last 100 years. More than 85 per cent of the times the monsoon has been normal, so those forecasts aren't really rocket science.

A forecast is only good if it can pick up early signals and warn of impending hazards so that policymakers and farmers in particular can prepare for the worst while hoping for the best. It is on this front that the long-range forecast made by IMD has failed. Last year, the IMD made a forecast that India will receive 'near-normal' rains with 96 per cent of the long period average expected, essentially numbing the government machinery. At the end of the season, the rains fell short by a whopping 22 per cent, leading to a shortfall of 12 per cent in paddy production, resulting in widespread food inflation.

The blame does not lie with IMD alone. Other global weather offices, including those from the US, Japan and the savvy private forecasters, also failed miserably. Unfortunately, the monsoon is still a scientific mystery, little understood despite huge advances.

Since our understanding of the

physics of the gigantic regional phenomenon called the monsoon is so poor, why does the Indian Government even try to predict the monsoon? It may be the right time for the scientists at IMD and other institutions to say 'we don't understand the physics of the tropical atmospheric chaos called the monsoon, we can't predict it'.

To make a point in their self-praise, IMD always points that from 1989 to 2000, they predicted a 'normal' monsoon and it 'turned out to be normal'. A closer examination reveals that in this 'golden period' their prediction was also way off the estimated margin of error allowed in the model itself. The flaw possibly lies in the statistical model that IMD uses to make its predictions; the current model is still not good enough, and a numerical model considered more accurate is still cooking.

Other experts also point out that there are continuous periods of time when the monsoon has behaved normally, showing little variation. They suggest that the 'golden 11-year period' was probably just another good epoch and nothing more. And since the IMD model tends to predict only a 'normal' monsoon, it came bang on target.

In fact, the litany of 'normal' monsoon only impedes early preparedness since ministries like agriculture, water resources and power that need to prepare for the possibility of extreme events are invariably caught unawares in part, thanks to the so-called scientific forecast by IMD. As a consequence, in bad years the Government has always had to scramble to meet the needs on an emergency basis.

To IMD's credit one must say that they are the only agency in the entire world that 'operationally predicts' the monsoon, all other institutions, in India and globally, do so more as an academic exercise. It is also widely acknowledged that even though the monsoon is a repetitive annual event, but to accurately predict the natural variations is like finding a path through a live minefield because the chaos of the tropical oceans and atmosphere is very hard to understand, let alone model and forecast. ■

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